Driver Coaching Report

Mode: vs_reference_session

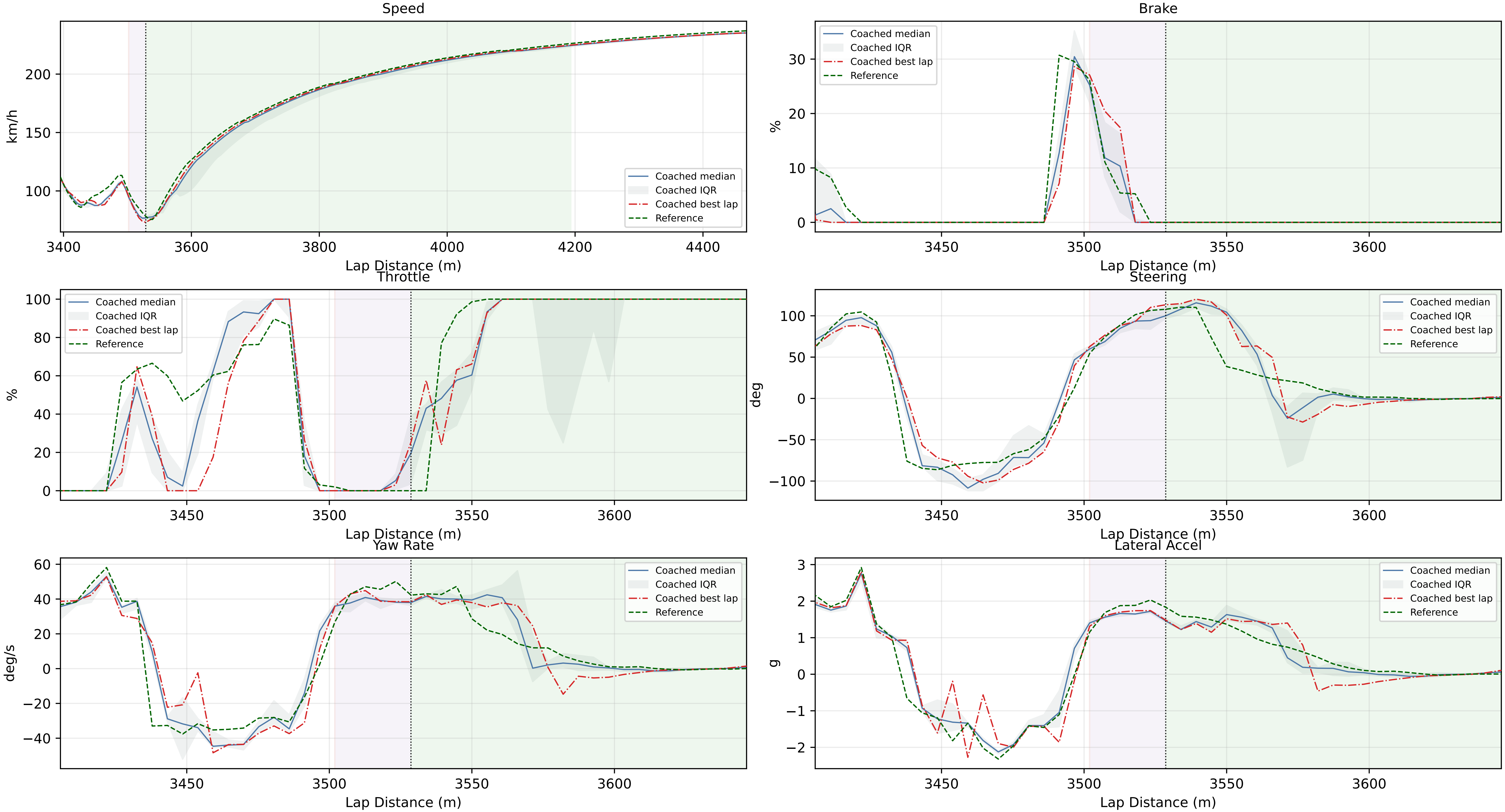
Session Context:
- Circuit: miami gp
- Session Type: Test
- Date/Time: 06/05/2026 17:34:40
- Sample Rate: 360, Hz
- Session Duration: 834.483, s
- Coached Driver: Unai Orbe Deusto
- Coached Car: superformulalights324
- Laps Used: 6 valid / 8 total
- Coached Best Lap: 111.297 s
- Coached Theoretical Optimal Lap: 109.196 s
- Coached Potential Gain (Best - Optimal): 2.101 s
- Corner Model: detected 19 corners
- Reference Driver: Unai Orbe Deusto
- Reference Car: superformulalights324
- Reference Best Lap: 110.447 s
- Reference Theoretical Optimal Lap: 107.699 s
- Reference Potential Gain (Best - Optimal): 2.748 s
- Pairing: Unai Orbe Deusto (coached) vs Unai Orbe Deusto (reference)

Main Entry Issue: No dominant entry issue in current priority set.
Main Mid-Corner Issue: Steering demand is too high through rotation.
Main Exit Issue: Exit phase is costing time onto the following straight.
Track Usage Assessment: Track-position proxy active (apex delta vs reference).

Graph Guide:
- Orange background = braking phase.
- Purple background = rotation / mid-corner phase.
- Green background = traction / exit phase.
- Vertical dotted line = apex point.
- Coached median = median trace across valid laps.
- Coached IQR = interquartile range (25th to 75th percentile) across valid laps.
- Coached best lap (red dash-dot) = fastest corner segment achieved by the coached driver.
- Reference = green dashed in vs-reference mode, red dashed in single-session mode.
- Speed uses a wider context window to show entry and exit consequence.
- Brake, throttle, steering, yaw-rate, and lateral-accel panels share one zoomed detail x-axis.
- X axis uses lap distance in meters when LapDist is available.

Top 3 Priorities:
1. T16 [exit]
   Key Issue: Exit phase is costing time onto the following straight.
   Priority Action: Commit throttle as steering opens, then build to full throttle with one smooth ramp.
   Practice Focus: 3 laps: repeat the full-throttle point within +/-5 m with no second steering correction.
   Confidence: medium
2. T17 [exit]
   Key Issue: Exit acceleration is weaker than reference.
   Priority Action: Prioritize cleaner car placement at apex so throttle can be committed earlier and harder. Apex placement also differs from reference, so prioritize a repeatable entry-to-apex path.
   Practice Focus: Focus on one corner only and maximize clean full-throttle point repeatability.
   Confidence: high
3. T11 [mid]
   Key Issue: Steering demand is too high through rotation.
   Priority Action: Reduce steering rate at turn-in and commit to one smooth arc through apex.
   Practice Focus: One-lap reset between attempts, focus on avoiding second steering input.
   Confidence: high

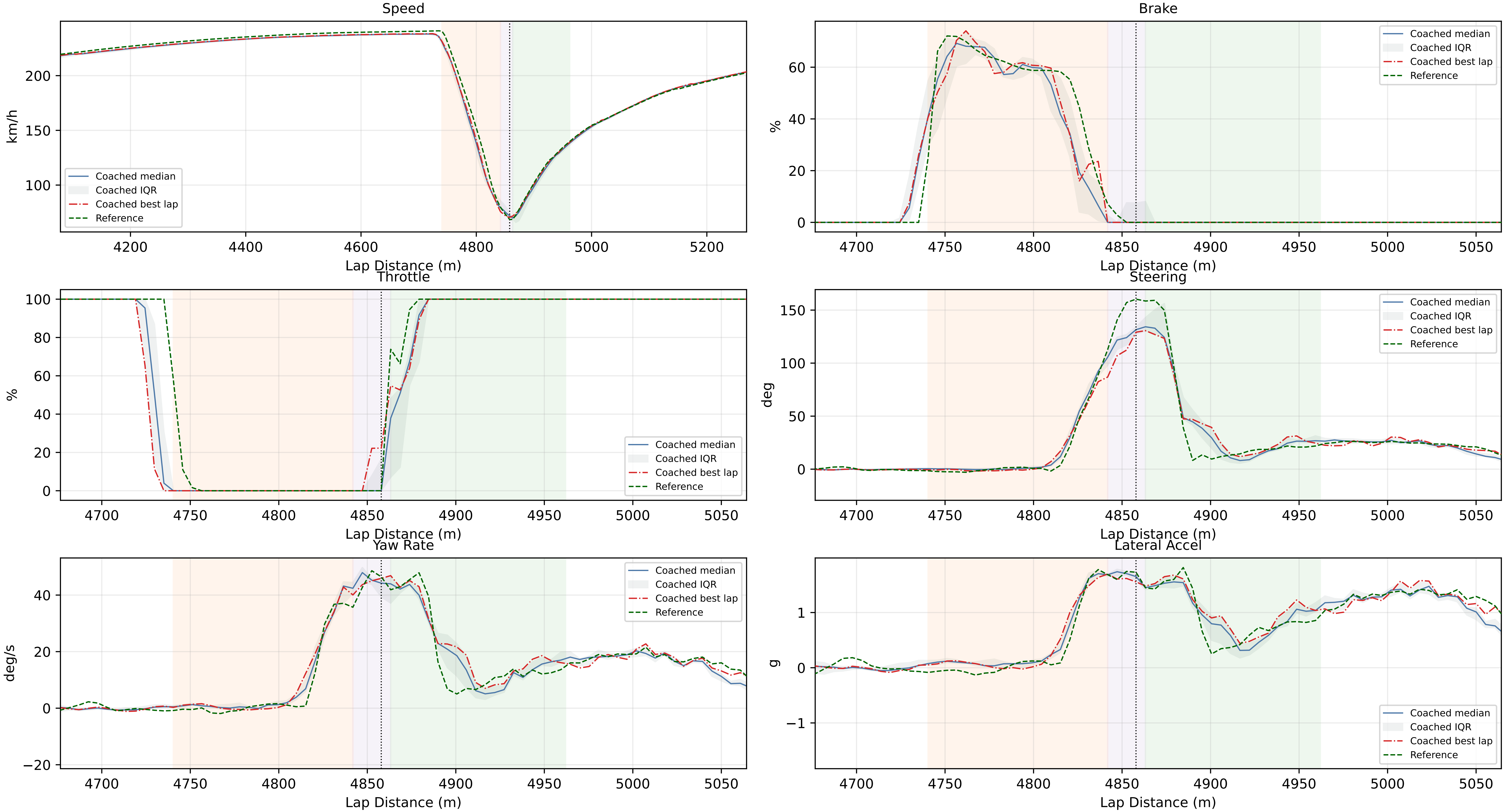
T16 Coaching (Priority 1)



Reference Segment Used: Reference driver lap 11
Coached Segment Used: Coached best lap 5

Coaching Summary: T16 | Exit priority. Symptom: Exit phase is costing time onto the following straight. Likely cause: Throttle commitment and steering unwind are not translating into early acceleration. Evidence: time loss +1.170 s; variability +1.899 s; apex speed delta +1.56 km/h; traction time delta +1.090 s; throttle reapply delay +0.000 % lap; exit long accel delta +0.004 g; apex position offset +1.43 m. Recommended action: Commit throttle as steering opens, then build to full throttle with one smooth ramp. Practice focus: 3 laps: repeat the full-throttle point within +/-5 m with no second steering correction. Confidence: medium (0.68).

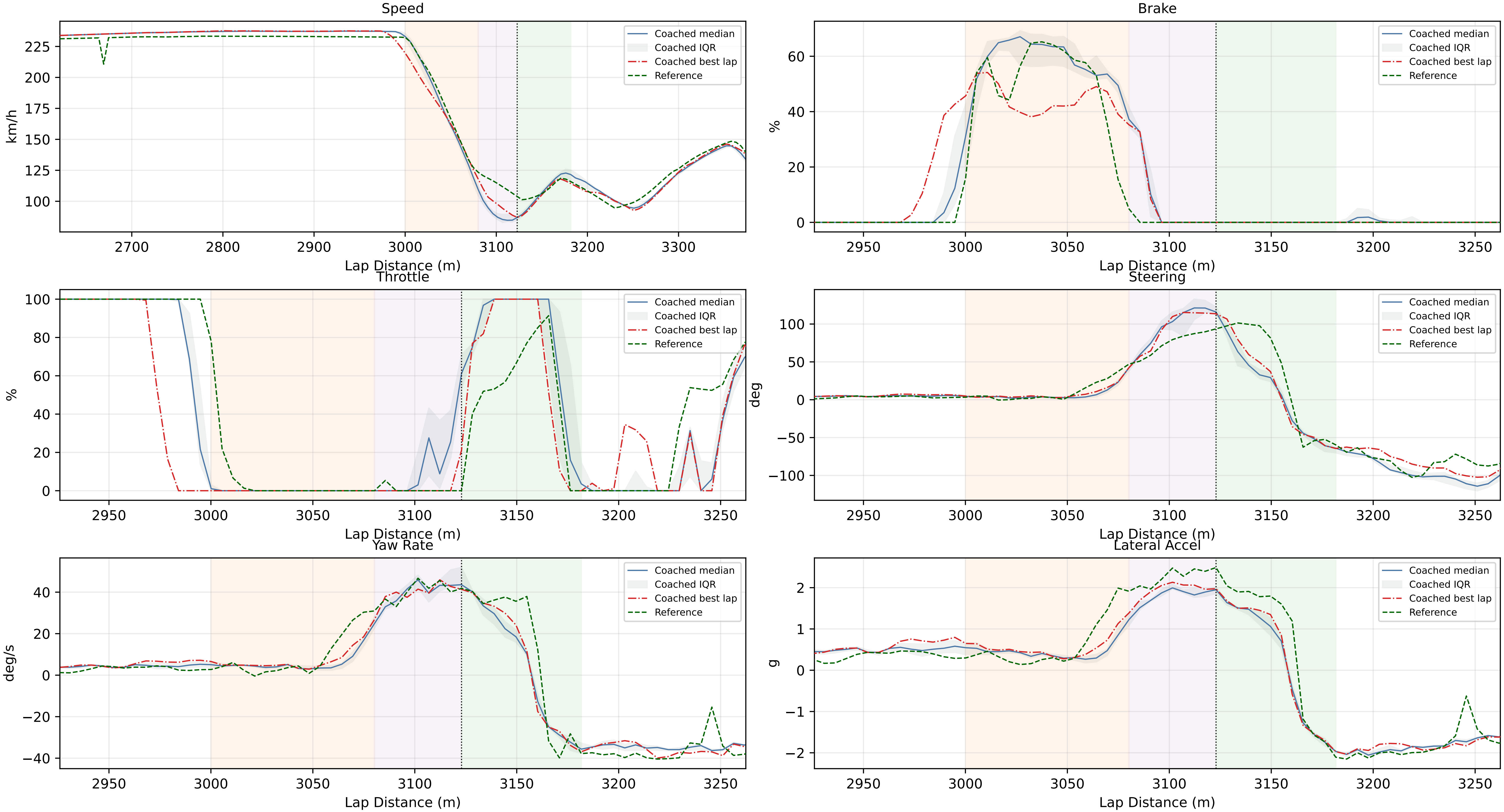
T17 Coaching (Priority 2)



Reference Segment Used: Reference driver lap 15
Coached Segment Used: Coached best lap 2

Coaching Summary: T17 | Exit priority. Symptom: Exit acceleration is weaker than reference. Likely cause: Longitudinal acceleration is below reference through traction phase. Positioning proxy indicates measurable apex-path offset versus reference. Evidence: time loss +0.731 s; variability +1.043 s; apex speed delta -3.80 km/h; traction time delta +0.288 s; throttle reapply delay +0.000 % lap; exit long accel delta -0.060 g; apex position offset +3.21 m. Recommended action: Prioritize cleaner car placement at apex so throttle can be committed earlier and harder. Apex placement also differs from reference, so prioritize a repeatable entry-to-apex path. Practice focus: Focus on one corner only and maximize clean full-throttle point repeatability. Confidence: high (1.00).

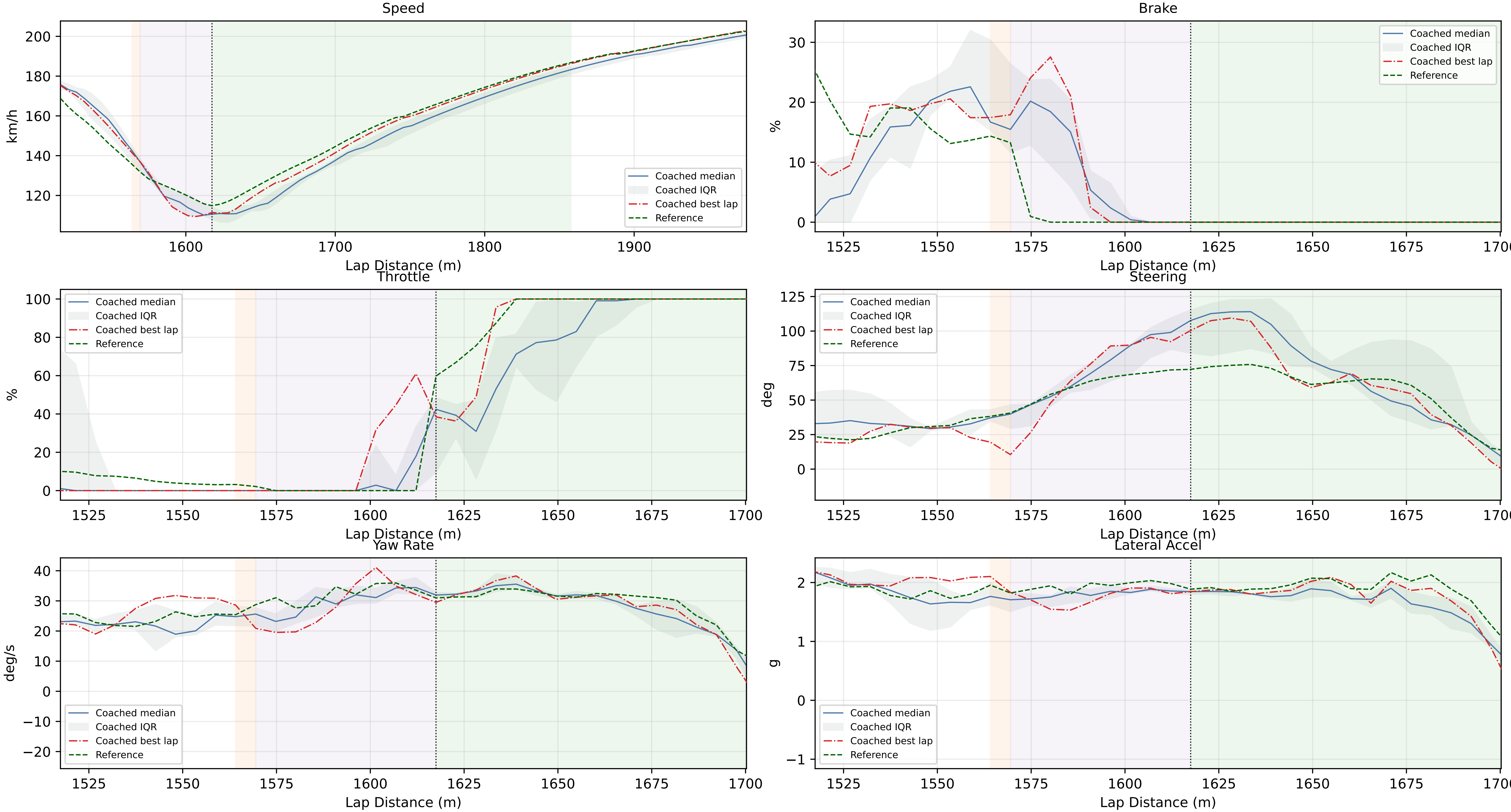
T11 Coaching (Priority 3)



Reference Segment Used: Reference driver lap 17
Coached Segment Used: Coached best lap 4

Coaching Summary: T11 | Mid-corner priority. Symptom: Steering demand is too high through rotation. Likely cause: Extra steering lock with lower apex speed suggests the car is being asked to turn too late. Evidence: time loss +0.569 s; variability +0.513 s; apex speed delta +17.24 km/h; rotation time delta +0.380 s; yaw delta +0.19 deg/s; steering delta +19.93 deg; apex position offset +0.67 m. Recommended action: Reduce steering rate at turn-in and commit to one smooth arc through apex. Practice focus: One-lap reset between attempts, focus on avoiding second steering input. Confidence: high (1.00).

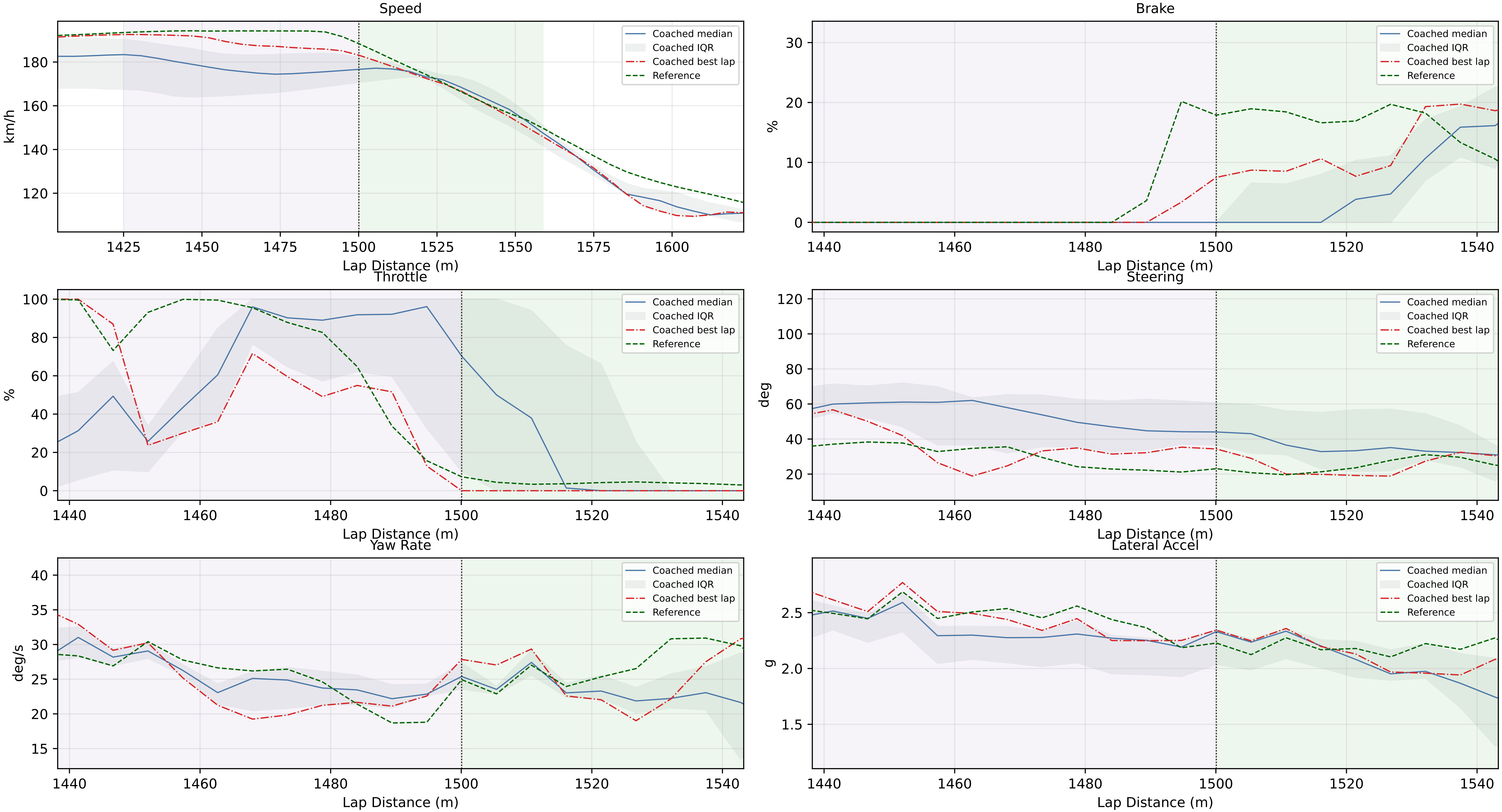
T8 Coaching (Priority 4)



Reference Segment Used: Reference driver lap 15
Coached Segment Used: Coached best lap 5

Coaching Summary: T8 | Exit priority. Symptom: Exit phase is costing time onto the following straight. Likely cause: Throttle commitment and steering unwind are not translating into early acceleration. Evidence: time loss +0.508 s; variability +0.463 s; apex speed delta +3.96 km/h; traction time delta +0.446 s; throttle reapply delay +0.000 % lap; exit long accel delta -0.028 g; apex position offset +1.04 m. Recommended action: Commit throttle as steering opens, then build to full throttle with one smooth ramp. Practice focus: 3 laps: repeat the full-throttle point within +/-5 m with no second steering correction. Confidence: medium (0.71).

T7 Coaching (Priority 5)



Reference Segment Used: Reference driver lap 17
Coached Segment Used: Coached best lap 5

Coaching Summary: T7 | Mid-corner priority. Symptom: Steering demand is too high through rotation. Likely cause: Extra steering lock with lower apex speed suggests the car is being asked to turn too late. Evidence: time loss +0.316 s; variability +0.389 s; apex speed delta +17.35 km/h; rotation time delta +0.223 s; yaw delta +0.18 deg/s; steering delta +23.72 deg; apex position offset +1.22 m. Recommended action: Reduce steering rate at turn-in and commit to one smooth arc through apex. Practice focus: One-lap reset between attempts, focus on avoiding second steering input. Confidence: high (1.00).